

ASSIGNMENT – 15.3

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Task Description #1 – Basic REST API Setup

Task: Ask AI to generate a Flask REST API with one route:

GET /hello → returns {"message": "Hello, AI Coding!"}

PROMPT:

Generate a Flask REST API with one route:

GET /hello → returns {"message": "Hello, AI Coding!"}

CODE:

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/hello', methods=['GET'])
def hello():
    return jsonify({"message": "Hello, AI Coding!"})

if __name__ == "__main__":
    app.run(debug=True)
```

```
* Serving Flask app '__main__'
* Debug mode: on
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
INFO:werkzeug:Press CTRL+C to quit
INFO:werkzeug: * Restarting with watchdog (inotify)
```

```
pip install flask
```

```
Requirement already satisfied: flask in /usr/local/lib/python3.12/dist-packages (3.1.2)
Requirement already satisfied: blinker>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in /usr/local/lib/python3.12/dist-packages (from flask) (8.3.0)
Requirement already satisfied: itsdangerous>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in /usr/local/lib/python3.12/dist-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from flask) (3.1.3)
```

```
[7] ✓ Os !python app.py
python3: can't open file '/content/app.py': [Errno 2] No such file or directory
```

```
[9] ✓ Os import requests

try:
    response = requests.get('http://127.0.0.1:5000/hello')
    print(response.json())
except requests.exceptions.ConnectionError as e:
    print(f"Error connecting to the Flask app. Make sure the Flask app is running in another cell. Error: {e}")

Error connecting to the Flask app. Make sure the Flask app is running in another cell. Error: HTTPConnectionPool(host='127.0.0.1', port=5000): Max retries exceeded with url:
```

```
[11] ✓ Os {"message": "Hello, AI Coding!"}

{"message": "Hello, AI Coding!"}
```

OBSERVATION:

I have observed that flask rest api is generated by ai and created a message hello, ai coding.

Task Description #2 – CRUD Operations (Students API)

Task:

Use AI to build REST endpoints for a Student API:

- GET /students → List all students.
- POST /students → Add a new student.
- PUT /students/<id> → Update student details.
- DELETE /students/<id> → Delete a student.

Expected Output:

- Flask API with dictionary/list storage.
- JSON responses for each operation

Prompt:

Generate GET /students → List all students
POST /students → Add a new student.
PUT /students/<id> → Update student details.
DELETE /students/<id> → Delete a student

Code:

```
# Install Flask
!pip install flask --quiet

from flask import Flask, request, jsonify

app = Flask(__name__)

# In-memory students database
students = [
    {"id": 1, "name": "Alice", "age": 20},
    {"id": 2, "name": "Bob", "age": 22}
]

# --- CRUD Endpoints ---

# GET /students
@app.route("/students", methods=["GET"])
def get_students():
    return jsonify(students)

# POST /students
@app.route("/students", methods=["POST"])
def add_student():
    data = request.get_json()
    if not data or "name" not in data or "age" not in data:
        return jsonify({"error": "Invalid data"}), 400
    new_id = students[-1]["id"] + 1 if students else 1
    student = {"id": new_id, "name": data["name"], "age": data["age"]}
```

```

        return jsonify({"error": "Invalid data"}), 400
    new_id = students[-1]["id"] + 1 if students else 1
    student = {"id": new_id, "name": data["name"], "age": data["age"]}
    students.append(student)
    return jsonify(student), 201

# PUT /students/<id>
@app.route("/students/<int:id>", methods=["PUT"])
def update_student(id):
    student = next((s for s in students if s["id"] == id), None)
    if not student:
        return jsonify({"error": "Student not found"}), 404
    data = request.get_json()
    student["name"] = data.get("name", student["name"])
    student["age"] = data.get("age", student["age"])
    return jsonify(student)

# DELETE /students/<id>
@app.route("/students/<int:id>", methods=["DELETE"])
def delete_student(id):
    global students
    student = next((s for s in students if s["id"] == id), None)
    if not student:
        return jsonify({"error": "Student not found"}), 404
    students = [s for s in students if s["id"] != id]
    return jsonify({"message": f"Student {id} deleted"})

# Run the Flask app in Colab
from threading import Thread

```

```

thread = Thread(target=run_app)
thread.start()

```

```

* Serving Flask app '__main__'
* Debug mode: off
Address already in use
Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.

```

```

import requests

BASE_URL = "http://127.0.0.1:5000/students"

# GET all students
print(requests.get(BASE_URL).json())

# POST a new student
print(requests.post(BASE_URL, json={"name": "Charlie", "age": 23}).json())

# PUT update student
print(requests.put(BASE_URL + "/2", json={"name": "Bobby", "age": 24}).json())

# DELETE a student
print(requests.delete(BASE_URL + "/1").json())

```

```

INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "GET /students HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "POST /students HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "PUT /students/2 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "DELETE /students/1 HTTP/1.1" 200 -

```

```
import requests

BASE_URL = "http://127.0.0.1:5000/students"

# GET all students
print(requests.get(BASE_URL).json())

# POST a new student
print(requests.post(BASE_URL, json={"name": "Charlie", "age": 23}).json())

# PUT update student
print(requests.put(BASE_URL+"/2", json={"name": "Bobby", "age": 24}).json())

# DELETE a student
print(requests.delete(BASE_URL+"/1").json())
```

```
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "GET /students HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "POST /students HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "PUT /students/2 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [04/Nov/2025 09:10:54] "DELETE /students/1 HTTP/1.1" 200 -
[{'age': 20, 'id': 1, 'name': 'Alice'}, {'age': 22, 'id': 2, 'name': 'Bob'}]
{'age': 23, 'id': 3, 'name': 'Charlie'}
{'age': 24, 'id': 2, 'name': 'Bobby'}
{'message': 'Student 1 deleted'}
```

[+ Code](#) [+ Text](#)

Observation:

This code creates an in-memory **Students REST API** in Colab using Flask, supporting GET, POST, PUT, and DELETE operations. It runs locally without ngrok, letting you test all endpoints directly in Colab.

Task Description #3 – API with Query Parameters

Task: Ask AI to generate a REST API endpoint

Expected Output:

Working search function with query param handling

Prompt: generate a REST API endpoint

Code:

REST

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash:generateContent" -H "x-goog-api-key: $GEMINI_API_KEY" -H 'Content-Type: application/json' -X POST -d '{
  "contents": [
    {
      "parts": [
        {
          "text": "Explain how AI works in a few words"
        }
      ]
    }
  ]
}'
```

API Keys

API quickstart Create API key

Group by API key Project Filter by All projects

Key	Project	Created on	Quota tier
...jYyY hello	Default Gemini Project gen-lang-client-0895986900	Nov 4, 2025	Set up billing Free tier
...OYP8 Default Gemini API Key	Default Gemini Project gen-lang-client-0895986900	Nov 4, 2025	Set up billing Free tier
...OYP8 Default Gemini API Key	Default Gemini Project gen-lang-client-0895986900	Nov 4, 2025	Set up billing Free tier

Can't find your API keys here?

This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys. You can also create a new API Key above.

Learn more

Task Description #4 – Integration & Testing

Task: Ask AI to write test scripts using Python requests module to call APIs created above.

Prompt:

Generate scripts using Python requests module to call APIs created above.

Code:

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/hello', methods=['GET'])
def hello():
    return jsonify({"message": "Hello, AI Coding!"})

# This part should ideally be run in a separate cell or in a way that keeps the server running.
# If run in the same cell without a mechanism to keep it alive (like a notebook environment),
# the cell will finish execution and the server will stop before the test request is sent.
# For demonstration purposes in a notebook where a separate persistent process is not easily shown,
# I will provide the test code in the next step.
# if __name__ == "__main__":
#     app.run(debug=True)
```

```
base_url = "http://127.0.0.1:5000"
try:
    response = requests.get(f"{base_url}/hello")
    print(f"Status Code: {response.status_code}")
    print("Response Body:", response.json())
except requests.exceptions.ConnectionError as e:
    print(f"Error connecting to the Flask app. Make sure the Flask app is running in another cell or pr

Error connecting to the Flask app. Make sure the Flask app is running in another cell or process. Error:
```

Write post request test

```
base_url = "http://127.0.0.1:5000"
new_student_data = {
    "name": "Charlie",
    "age": 25
}

try:
    response = requests.post(f"{base_url}/students", json=new_student_data)
    print(f"Status Code: {response.status_code}")
    print("Response Body:", response.json())
except requests.exceptions.ConnectionError as e:
    print(f"Error connecting to the Flask app. Make sure the Flask app is running in another cell or pr

Error connecting to the Flask app. Make sure the Flask app is running in another cell or process. Error:
```

Write put request test

```
base_url = "http://127.0.0.1:5000"
student_id_to_delete = 1 # Assuming student with ID 1 exists

try:
    response = requests.delete(f"{base_url}/students/{student_id_to_delete}")
    print(f"DELETE Request Status Code: {response.status_code}")
    print("DELETE Request Response Body:", response.json())
except requests.exceptions.ConnectionError as e:
    print(f"Error connecting to the Flask app. Make sure the Flask app is running in another cell or pr")
except Exception as e:
    print(f"An unexpected error occurred: {e}")
```

Error connecting to the Flask app. Make sure the Flask app is running in another cell or process. Error:

Summarize results

Observation :

Attempts to test the API endpoints with requests resulted in `ConnectionError`. This indicates the Flask application is not running or accessible. The tests for GET, POST, PUT, and DELETE all failed. To run the tests successfully, the Flask app must be active. Ensure the cell running the Flask server is executed before the test cells.